

SMART CONTRACT

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- Introduction to Smart contracts, usage, application, working principle, Law and Regulations

SMART CONTRACT

- A smart contract is an electronic transaction protocol that executes the terms of a contract. The general objectives are to satisfy common contractual conditions, minimize exceptions both malicious and accidental, and minimize the need for trusted intermediaries.
- A smart contract is a secure and unstoppable computer program representing an agreement that is automatically executable and enforceable.
- A Smart Contract (or cryptocontract) is a computer program that directly and automatically controls the transfer of digital assets between the parties under certain conditions.

History

- In 1994, Nick Szabo, a legal scholar, and a cryptographer recognized the application of a decentralized ledger for smart contracts.
- He theorized that these contracts could be written in code which can be stored and replicated on the system and supervised by the network of computers that constitute the blockchain.
- These smart contracts could also help in transferring digital assets between the parties under certain conditions.

Properties

- Automatically executable
- Distributed
- Enforceable
- Customizable
- Trustless
- Secure
- Deterministic
- Semantically sound
- Unstoppable

Capabilities of Smart Contracts

- Accuracy
- Automation
- Speed
- Backup
- Security
- Savings
- Manages information
- Multi-signature accounts

How does a Smart Contract Work?



Identify Agreement

Multiple parties identify the cooperative opportunity and desired outcomes.



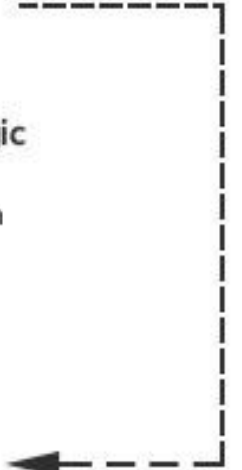
Set conditions

Smart contracts are executed automatically when certain conditions are met.



Code business logic

A computer program is written



Encryption and blockchain technology

Encryption provides a secure transfer of messages between parties.



Execution and processing

The code is executed and outcomes are memorialized.



Network updates

All the nodes on the network update their ledger.

Steps needed for smart contract

- Agreement
- Contract creation
- Deployment
- Monitoring conditions
- Execution
- Recording

Types of Smart Contract

- Smart legal contract
- Decentralized Autonomous Organization (DAO)
- Application Logic Contracts

Advantages of smart contract

- Recordkeeping
- Autonomy
- Reduce fraud
- Fault tolerance
- Enhanced trust
- Cost – efficiency

Challenges of smart Contract

- No regulations
- Difficult to implement
- Immutable
- Alignment